

Introduction to bioengineering

Instructor: Dr Ayan Sadhukhan

TAs: Debankona, Surbhi, Nityanand, Amlan,
Revanth, Harita, Rabisankar, Amandeep

Class code: ls3qlvsm

Classes and evaluation pattern

Slides and videos

Continuous evaluation:

Presentation and assignments: 20 marks

Attendance and class participation: 10 marks

Minor exam: 30 marks

Major exam: 40 marks

Other information

- **Class Schedule:** Monday, Wednesday, Friday @ 8-9 AM
- Please come to the class on time. The classroom doors will be closed by 8.10 PM and attendance will not be counted thereafter. Seat according to the plan.
- **Dates to remember:**
 - Minor 1 Examination: 17-20 September 2025
 - Semester break: 18-26 October 2025
 - Major Examination: 20 – 26 Nov 2025

Nature's curiosities

Tardigrades (water bear)
size upto 1.2mm



Drosophila melanogaster
size ~ 3 mm



Protopterus (megabats)
size ~1.5 m



Marine mussel
size 5cm- 6cm



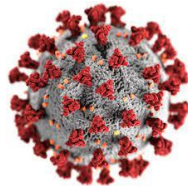
Barn owl
size ~30 - 40 cm



Stenocara gracilipes (Namib beetle)
size range ~ 3 cm



African bush elephant
Size ~3 m



Coronavirus
Size ~ 0.0007mm

Get Curious



0.3 to 0.5 mm in length

Only 1000 cells

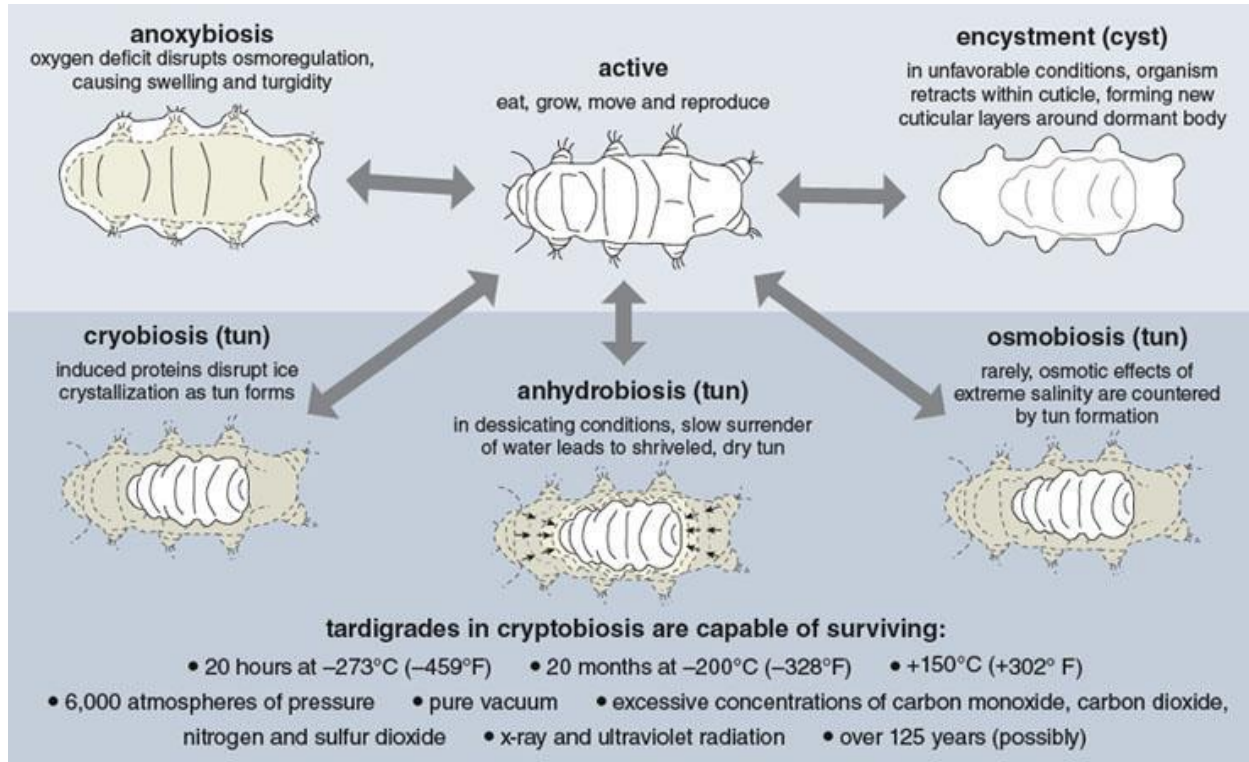
Can lose 95 % of the water and go in suspended animation

Can survive even upto 30 years without food and water

Can live in absolute zero temp, volcanos, ocean's trenches and in space

Can survive intense ionizing radiations

Nature's innovations for today's solutions



Inspirations

Storage

Cold chain for vaccination

Avoiding damage in radiation

AMERICAN SCIENTIST BY [WILLIAM RANDOLPH MILLER](#) 2011
VOLUME 99, NUMBER 5

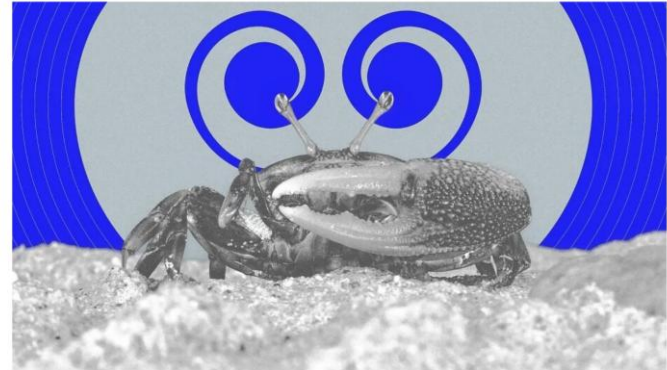
Next-generation cameras inspired by fruit flies, moths and crabs



08-11-22 | 10:00 AM

MIT scientists have developed a new kind of camera—from a crab

Crabs see as well underwater as over it. This new camera does, too.



[Source photo: lessydoang/Getty Images]

artificial compound eyes that could make camera lenses tiny enough to fit on credit cards or stickers, autonomous vehicle, drones

Problem with Smart Glues



- Contains Formaldehyde – Carcinogen
- Irreversible
- Ineffective under water

Marine mussel



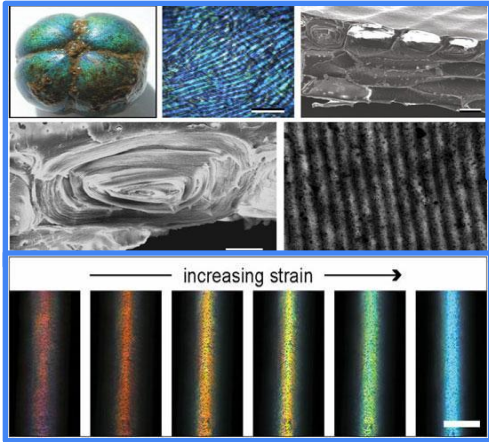
Underwater adhesion ?

adhesive protein that contains a unique amino acid (DOPA) with a catechol side chain responsible for strong adhesion to underwater surface

Current /pH change can modify catechol

Smart underwater adhesive

From Basics to Bio-inspired Designs



Color Changing fibers: Inspired from hogberry



Elephant inspired Bionic Handling



Woodpecker inspired
Extreme Shock Absorber

Paint without Pigments: Inspired from peacock feathers



From Basics to Solving problems



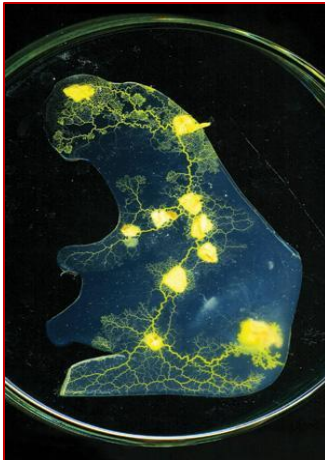
Fog Harvesting system:

Inspired from Namib beetle



Chemical-Free De-Icing:

Inspired from water strider



Traffic Decongestant:

Inspired from slime mould's path making tactic



Less Noisy, efficient Fan:

Inspired from maple tree seedpod wings

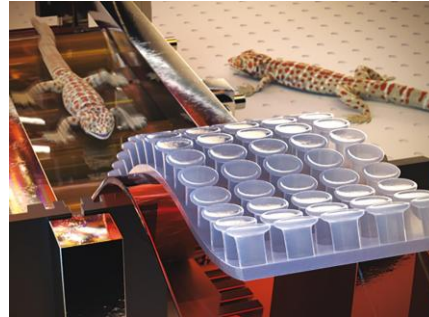


Living systems as Natural Bioengineers: Source of Bioinspired innovations

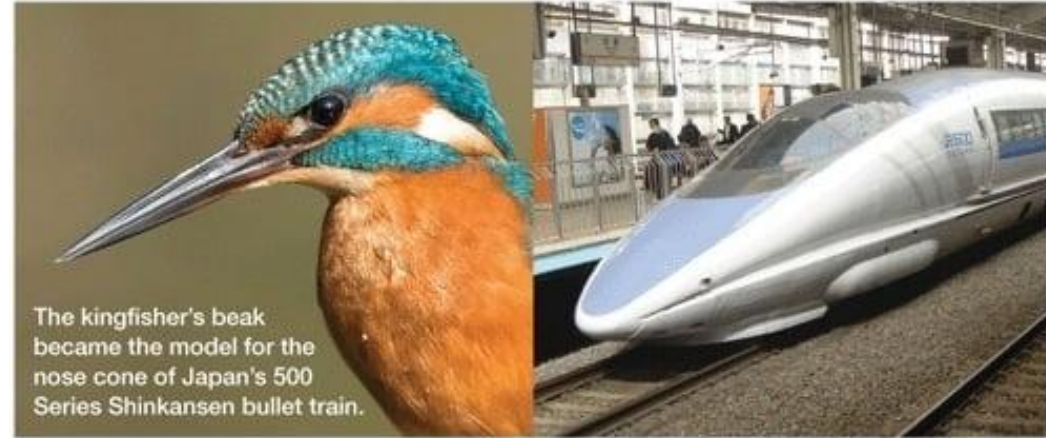


Velcro from Burr

Adhesive from Gecko



The Shinkansen Bullet Train Inspired by the kingfisher



The kingfisher's beak became the model for the nose cone of Japan's 500 Series Shinkansen bullet train.

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*Curiosity of natural systems
can spur innovative solutions*

Engineering marvels



Box fish to Nextgen Merc- Bionic



Termite houses to AC free buildings

Inspiration from spiders for Mission mars

Realization of rolling locomotion by a wheel-spider-inspired hexapod robot

[Takuma Nemoto](#) ✉, [Rajesh Elara Mohan](#) & [Masami Iwase](#)

[Robotics and Biomimetics](#) **2**, Article number: 3 (2015) | [Cite this article](#)

2515 Accesses | **3** Citations | [Metrics](#)

Abstract

This study aims to realize rolling locomotion by a multi-legged robot inspired by “wheel spider” in nature. A novel robot mechanism and a strategy for rolling locomotion is proposed in this paper based on motion analysis of wheel spider species in nature. We also present details of a model of a wheel-spider-inspired hexapod robot and design of its controller in realizing rolling locomotion. Results of numerical simulations validated the efficacy of the proposed approach in synthesizing rolling locomotion in a wheel-spider-inspired hexapod robot.



Cart wheeling spider

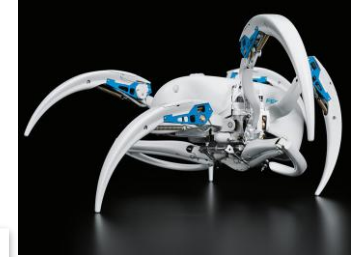


Image of mars
acquired by mars
rover perseverance



Biological solutions and limitless possibilities



Bombardier beetles have the infamous ability to synthesize and release rapid bursts of stinky, burning-hot liquid from their rear ends



By plugging the fungi into a synthesiser, the **mushrooms make noises** which are comparable to trance music due to its pitch, pace and distinct hypnotic sound. The bioelectrical impulses are converted into sounds

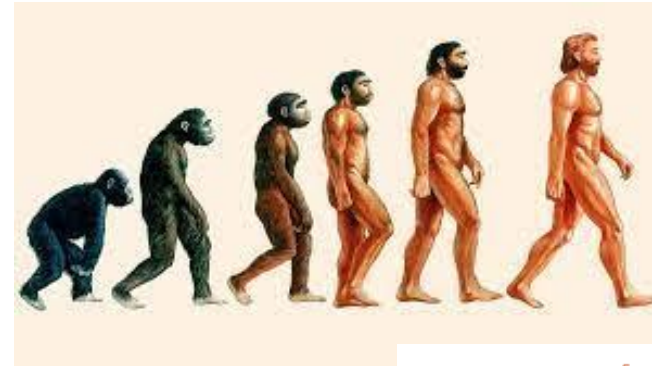
Biological insights from intelligent Genomes

Naked mole rat : cancer & certain pains

Python : dynamic increase in organ mass



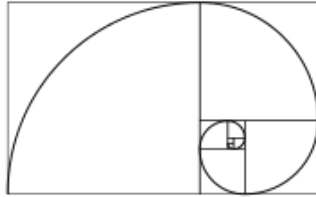
Oyster : Survival strategies



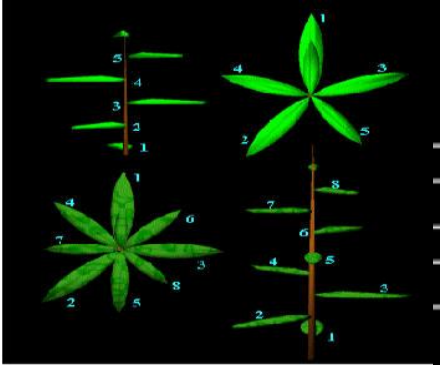
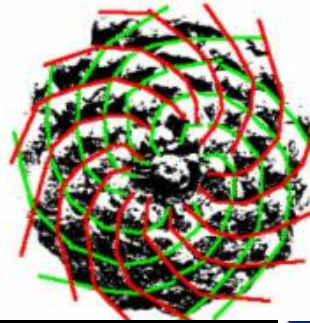
Observe Nature

Fibonacci series and the golden ratio

1, 1, 2, 3, 5, 8, 13, 21, 34, 55, 89, 144, ...



An approximation of the golden spiral created by drawing circular arcs connecting the opposite corners of squares in the Fibonacci tiling; this one uses squares of sizes 1, 1, 2, 3, 5, 8, 13, 21, and 34.



More robotics inspirations from nature

<https://asknature.org/biological-strategies/>

~ 1700 biological strategies; 57 innovations

ask nature

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